- Your application may not handle edge cases properly and may instead crash unexpectedly.

Regardless of whether the host system or your application is the cause of a failure, a single EC2 instance is a single point of failure. If you rely on a single EC2 instance, your system will fail up eventually. It's merely a matter of time.

16.1.1 Redundancy can remove a single point of failure

Imagine a production line that makes fluffy cloud pies. Producing a fluffy cloud pie requires the following production steps (simplified!):

1. Produce a pie crust.
2. Cool the pie crust.
3. Put the fluffy cloud mass on top of the pie crust.
4. Cool the fluffy cloud pie.
5. Package the fluffy cloud pie.

The current setup is a single production line. The big problem with this process is that whenever one of the steps crashes, the entire production line must be stopped. Figure 16.1 illustrates what happens when the second step (cooling the pie crust) crashes. The steps that follow no longer work, because they no longer receive cool pie crusts.